TITLE SLIDE

CardioTwin AI: Virtual Cardiovascular Patient Laboratory

Digital Twin Challenge 2026 organized by Happiest Health

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- Core Focus: In-silico cardiovascular digital twins fusing EHR & wearable telemetry
- License: MIT Open-Source License • Local Zero-Cloud Execution

CLINICAL BACKGROUND

Problem Statement: The Post-Discharge Monitoring Blindspot

Fragmented episodic clinic visits vs continuous home deterioration

- Cardiovascular diseases (CVDs) remain the leading cause of global mortality.
- Traditional healthcare relies on episodic 15-minute consultations every 3 to 6 months.
- Acute autonomic destabilization often transpires within hours at home without warning.
- Ordinary consumer wearables record data, but lack clinical context from baseline EHR records.
- Need: A personalized computational avatar that fuses static clinical history with dynamic physiological sensors.

CONCEPT & INNOVATION

Why a Cardiovascular Digital Twin?

Transforming passive telemetry into active predictive foresight

- Personalized Computational Avatar: Mirrors individual patient physiology in software.
- Continuous Multi-Modal Fusion: Combines static EHR risk factors with high-frequency wearable streams.
- Dynamic 24-Hour Horizon: Continuously updates deterioration risk when new readings arrive.
- Counterfactual 'What-If' Simulation: Allows clinicians to simulate interventions (e.g. sleep improvement, BP control) without patient harm.
- Auditable Decision Trail: Every inference and sensor reading is permanently logged.

CLINICAL SPECIFICATION

Target Healthcare Use Case & Prediction Scope

Synthesized 24-hour deterioration proxy under strictly causal lookback

- Target Definition: Composite physiological deterioration over the subsequent 24-hour window.
- Trigger Criteria: Sustained tachycardia (>=95 bpm), parasympathetic collapse (HRV RMSSD <=18 ms), or acute hypertensive surge (SBP >=165 mm
- Strict Zero-Leakage Policy: No future observations utilized in feature extraction at prediction time T.
- Scientific Honesty Disclaimer: Prototype evaluates a synthetic proxy; does not predict heart attacks or replace clinical judgment.
- Outcome Prevalence: 29.0% positive window occurrence across the 5,550 observation cohort.

TECHNICAL ARCHITECTURE

End-to-End System Architecture

Modular, maintainable, and completely reproducible local pipeline

- Data Pipeline: Synthetic EHR Generator + 14-day multi-frequency wearable telemetry simulator.
- Feature Engineering: Causal rolling temporal windows (6h mean HR, 24h min HRV, sleep debt, rate-pressure index).
- Machine Learning Engine: Tuned Random Forest (150 trees) vs Standardized Logistic Regression baseline.
- Persistence Layer: Embedded SQLite database logging patients, telemetry, predictions, and simulation events.
- Backend: High-performance Python FastAPI service with Pydantic schema validation.
- Frontend: Responsive React + TypeScript dashboard with Recharts and Tailwind CSS.

DATA ENGINEERING

Static Baseline EHR Ingestion

Anatomical and chronic vascular baseline parameters

- Demographics: Age (38-80 yrs), Biological Sex, Body Mass Index (BMI).
- Comorbidities: Clinically correlated Hypertension, Type 2 Diabetes, Smoking Status, Prior MI history.
- Hemodynamic & Lipid Markers: Baseline Systolic/Diastolic BP (mmHg), Total Cholesterol, HDL, LDL (mg/dL).
- Renal Filtration Markers: Serum Creatinine (mg/dL) and CKD-EPI estimated eGFR (mL/min).
- Provenance & Integrity: Seed 42 determinism, missingness indicators, and audit creation timestamps.

DATA ENGINEERING

Dynamic Wearable Telemetry Streams

Time-indexed physiological signals captured at 3-hour intervals

- Continuous Heart Rate: Instantaneous pulse rate in beats per minute (bpm).
- Heart Rate Variability (HRV): Root Mean Square of Successive Differences (RMSSD in ms) reflecting parasympathetic autonomic tone.
- Sleep Architecture: Daily nocturnal sleep duration (hours) and actigraphy sleep quality index (0-100).
- Physical Activity: Interval step count and ambulatory movement intensity.
- Ambulatory Blood Pressure: Physiologically correlated estimated systolic BP.
- Simulated Loss: Realistic 2.0% packet dropout rate with causal forward-fill imputation.

MACHINE LEARNING

Machine Learning Pipeline & Zero-Leakage Validation

Patient-level grouped holdout evaluation with zero cross-patient overlap

- Partitioning: GroupShuffleSplit on unique patient IDs (40 training patients, 10 held-out test patients).
- Zero Data Leakage: Verified 0 overlapping patients between train and test splits.
- StandardScaler Isolation: Preprocessing fitted exclusively on the 40 training patients.
- Baseline Model: Logistic Regression with L2 regularization and balanced class weights.
- Comparison Model: Random Forest Classifier (150 estimators, max_depth=6, min_samples_leaf=4).
- Explainability: Gini impurity ranking and local feature deviation vectors.

RESULTS & VALIDATION

Empirical Evaluation Results (Held-Out Test Cohort)

Genuine executed metrics from saved artifacts - zero fabricated numbers

- ROC-AUC: Random Forest achieved 0.9367 vs Logistic Regression 0.8217 (+11.5% gain).
- Average Precision (PR-AUC): Random Forest achieved 0.8939 vs Logistic Regression 0.8233.
- Balanced F1-Score: Random Forest achieved 0.8295 vs Logistic Regression 0.6733.
- Recall (Sensitivity): Random Forest detected 93.29% of deterioration windows (vs 85.53% LR).
- Brier Calibration Loss: 0.1221 (RF) vs 0.2373 (LR) demonstrating superior probability reliability.
- Confusion Matrix (N=1,110 test samples): TN=482, FP=151, FN=32, TP=445.

USER EXPERIENCE

Interactive Prototype: Research Dashboard & What-If Studio

Six connected screens built for clinical researchers

- Screen A (Overview): Cohort-wide risk distribution, active telemetry streams, and patient explorer.
- Screen B (Patient Twin): Static EHR baseline cards vs dynamic Recharts (HR, HRV RMSSD ms, Sleep, Steps).
- Screen C (Simulation & What-If): Timeline advancer, stress spike injection, and counterfactual parameter sliders.
- Screen D (Evaluation): Genuine ROC/PR metrics, confusion matrices, and calibration disclosure.
- Screen E (Explainability): Gini feature rankings, data quality monitors, and correlation vs causation alerts.
- Screen F (About): Competition background, student details, and ethical safeguards.

ETHICS & SAFETY

Safety, Ethical Safeguards & Medical Disclaimers

Scientific honesty and strict adherence to competition boundaries

- 100% Synthetic Data: Zero real patient records or identifiable health information (PHI) utilized.
- No Medical Overclaiming: Explicitly disclaims diagnosis, treatment prescription, or heart attack prediction.
- Correlation != Causation: High feature importance denotes algorithmic association, not biological proof.
- Prominent Dashboard Banners: Unmissable research prototype disclaimers on every screen.
- Local Confidentiality: No telemetry or patient IDs are ever uploaded to cloud or commercial APIs.

CONCLUSION & NEXT STEPS

Expected Impact, Future Roadmap & Conclusion

Bridging academic digital-twin prototypes with future clinical validation

- Clinical Trial Roadmap: Prospective multi-center validation with institutional review board (IRB) approval.
- Sensor Benchmark: Cross-validating wearable PPG sensors against gold-standard 12-lead Holter ECGs.
- Demographic Tuning: Calibrating distributions against representative Indian population cardiovascular registries.
- Regulatory Compliance: Building SaMD audit documentation compliant with CDSCO and FDA guidelines.
- Summary: CardioTwin AI delivers a fully working, reproducible, technically credible digital-twin prototype.